Rohith Gopinath

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Indian Institute Of Technology Bombay

Examination	University	Institute	Year	CPI / % Credits
M.Sc. in Physics	IIT Bombay	IIT Bombay	2027	8.99 Core: 73 | Total: 79
B.Sc. (Hons) in Physics	University Of Delhi	St Stephen's College	2024	8.8
High School	Kerala Board of Higher Secondary Education	St Pauls EMHSS Thenhipalam	2021	99.50%

ACADEMIC ACHIEVEMENTS

- **ALL INDIA RANK - 10, Joint Admission Test for Masters in Physics** (2025)
- **Erasmus Mundus Joint Master Scholarship, NUCPHYS** (2024)
- **INSPIRE Scholarship for Higher Education, Ministry of Science and Technology, Government of India** (2024)

RESEARCH AND PROJECTS

Master's Thesis, Indian Institute Of Technology Bombay • Dr. Sai Vinjanampathy (**2026 - Present**)

(Working title: Clifford-Augmented Matrix Product States for Quantum Reservoir Computing)

- Investigating Clifford-augmented matrix product-state representations for the simulation and analysis of quantum reservoir dynamics.
- Current work focuses on formulating the reservoir architecture, input-encoding protocol, readout observables, and numerical benchmarking strategy, informed by the literature on quantum reservoir computing, Clifford methods, and tensor-network simulation..

Summer Research Intern, Indiana University • Dr. Nicholas J. Mayhall (**2026**)

(Sparse Pauli Propagation and Transfer-Tensor Methods for Non-Markovian Quantum Dynamics)

- Developed a computational framework combining sparse Pauli propagation with the Transfer Tensor Method, using short-time Heisenberg-picture trajectories to construct subsystem dynamical maps and propagate observables to longer times.
- Implemented the workflow in Julia, including Pauli-subspace projection, transfer-tensor construction and propagation, memory and stability diagnostics, and validation against direct Pauli propagation and exact small-system references.
- Applied the method to spin-boson models; obtained agreement with direct references within numerical convergence tolerance in the tested low-coupling spin-boson regime, while analysing errors from Trotterization, sparse truncation, and finite memory windows.
- Results from this project are being developed toward a research manuscript with Prof. Nicholas J. Mayhall.

Semester Project, Indian Institute Of Technology Bombay • Dr. Soumya Bera (**2026**)

(Eigenstate Thermalization and Quantum Chaos in Quantum Spin Chains)

- Implemented exact diagonalization of a mixed-field Ising chain in Python, using bit-state encoding and resolving spin-flip and reflection symmetries to compare integrable and non-integrable regimes.
- Analysed unfolded level-spacing distributions and adjacent-gap ratios against Poisson and Gaussian orthogonal ensemble predictions, tracking the crossover to quantum-chaotic spectral statistics as the longitudinal field was varied.
- Investigated the eigenstate thermalization hypothesis through eigenstate expectation values of local magnetization and nearest-neighbour correlations, half-chain entanglement entropy, and the system-size dependence of eigenstate fluctuations.

Semester Project, Indian Institute Of Technology Bombay • Dr. Maniraj Mahalingam (**2026**)

(Design and Construction of an Ionic Plasma Thruster)

- Built an ionic plasma thruster from scratch as a team project, integrating the propulsion assembly with a custom high-voltage driving circuit.
- Constructed a zero-voltage-switching (ZVS) driver and integrated a voltage step-up stage to generate the high voltage required for operation from a conventional power supply.
- Demonstrated thrust generation with the completed prototype, providing a qualitative demonstration of ionic propulsion.

Summer Research Intern, IISER Thiruvananthapuram • Dr. Chandrakala Meena (**2024**)
(Nonlinear Dynamics of Attractor Systems)

- Received research training in qualitative and computational nonlinear dynamics, applying bifurcation analysis and initial-condition sensitivity studies to characterize periodic, quasiperiodic, and chaotic behaviour across 12 nonlinear strange-attractor systems.
- Implemented the dynamical systems in Python and systematically compared Euler, RK2, and RK4 integration schemes, assessing their numerical stability, computational efficiency, and ability to resolve transitions between distinct dynamical regimes.

Research Student, Department Of Mathematics, St Stephen's College, University of Delhi (**2024**)

• Advisor : *Dr. Piyush Bansal* • Topic : *Applications of Operator Theory*

- Undertook a guided study of the operator-theoretic and functional-analytic foundations of quantum mechanics, with emphasis on self-adjoint operators, spectral decomposition, and functional calculus.
- Applied these methods to Hamiltonian representations and canonical commutation relations, examining how spectral properties and operator structure determine the mathematical formulation and evolution of quantum systems. Prepared and presented the project report *Operator Theory and Applications*

CONFERENCES, PRESENTATIONS & TRAINING

56th Midwest Theoretical Chemistry Conference (MWTCC56) (2026)
(University of Minnesota, Minneapolis, USA)

- Attendee as a research intern with the Mayhall group at Indiana University.

4th International Conference on Quantum Information and Technology(QIQT) (2023)
(Indian Institute Of Science Education and Research, Kolkata, India)

- Participated in expert-led lectures and tutorials spanning quantum information theory, quantum optics, quantum chemistry, quantum foundations, and quantum materials and attended the Researchers' Meet.

Quantum Excellence Award (2023)

(Global Summer School On Quantum Computing, **IBM**)

- Completed training in quantum algorithms, circuit construction, and execution on IBM Quantum systems using the Qiskit SDK; received the Quantum Excellence Award for achieving the highest grade in a global cohort of more than 5,000 participants.

Meera Memorial Paper Presentation (2023)
(The Physics Society, St. Stephen's College)

- Presented a critical review titled *Objective Collapse Theories and Interpretations of Quantum Mechanics*, examining the quantum measurement problem and comparing the GRW, CSL, and Diósi–Penrose collapse models.

EXTRACURRICULARS

Founder, ClosedSpace™ (2025–Present)
Kerala,India

- Product development and business-model design for an early-stage wellness startup.

Teacher, Social Service League (2023-2024)
St Stephen's College, Delhi

- Taught, mentored and trained high school STEM students from underprivileged backgrounds in physics and mathematics under the society's evening classes forum.

TECHNICAL SKILLS

Programming Languages: Python, Julia, Fortran
Tools : Mathematica, Git, Linux, Bash, SLURM/HPC workflows, LaTeX
Research Interests : Quantum machine learning and reservoir computing;
Classical simulation of quantum systems, Tensor-network and Pauli-based methods;
Open quantum systems and Non-Markovian dynamics.
HOBBIES : Philosophy, Fitness, and Basketball.